

Natural Language Processing

Natural Language Processing

- Computers and machines are great at working with tabular data or spreadsheets.
- However, as human beings generally communicate in words and sentences, not in the form of tables.
- Much information that humans speak or write is unstructured. So it is not very clear for computers to interpret such.
- In natural language processing (NLP), the goal is to make computers understand the unstructured text and retrieve meaningful pieces of information from it.
- Natural language Processing (NLP) is a subfield of artificial intelligence, in which its depth involves the interactions between computers and humans.

Natural Language Processing

- NLP is a branch of data science that consists of systematic processes for analyzing, understanding, and deriving information from the text data in a smart and efficient manner.
- By utilizing NLP and its components, one can organize the massive chunks of text data, perform numerous automated tasks and solve a wide range of problems such as –
 - automatic summarization, machine translation, named entity recognition, relationship extraction, sentiment analysis, speech recognition, and topic segmentation etc.

Applications of NLP

- Machine Translation.
- Speech Recognition.
- Sentiment Analysis.
- Question Answering.
- Summarization of Text.
- Chatbot.
- Intelligent Systems.
- Text Classifications.
- Character Recognition.
- Spell Checking.
- Spam Detection.
- Autocomplete.
- Named Entity Recognition.
- Predictive Typing.

Understanding NLP

Example 1 :

I saw a man on a hill with a telescope.

These are some interpretations of the sentence shown above.

- There is a man on the hill, and I watched him with my telescope.
- There is a man on the hill, and he has a telescope.
- I'm on a hill, and I saw a man using my telescope.
- I'm on a hill, and I saw a man who has a telescope.
- There is a man on a hill, and I saw him something with my telescope.

Example 2 :

Can you help me with the can?

Understanding NLP

- Language processing is not “deterministic” (the same language has the same interpretations), and something suitable to one person might not be suitable to another.
- Therefore, Natural Language Processing (NLP) has a non-deterministic approach.
- In other words, Natural Language Processing can be used to create a new intelligent system that can understand how humans understand and interpret language in different situations.

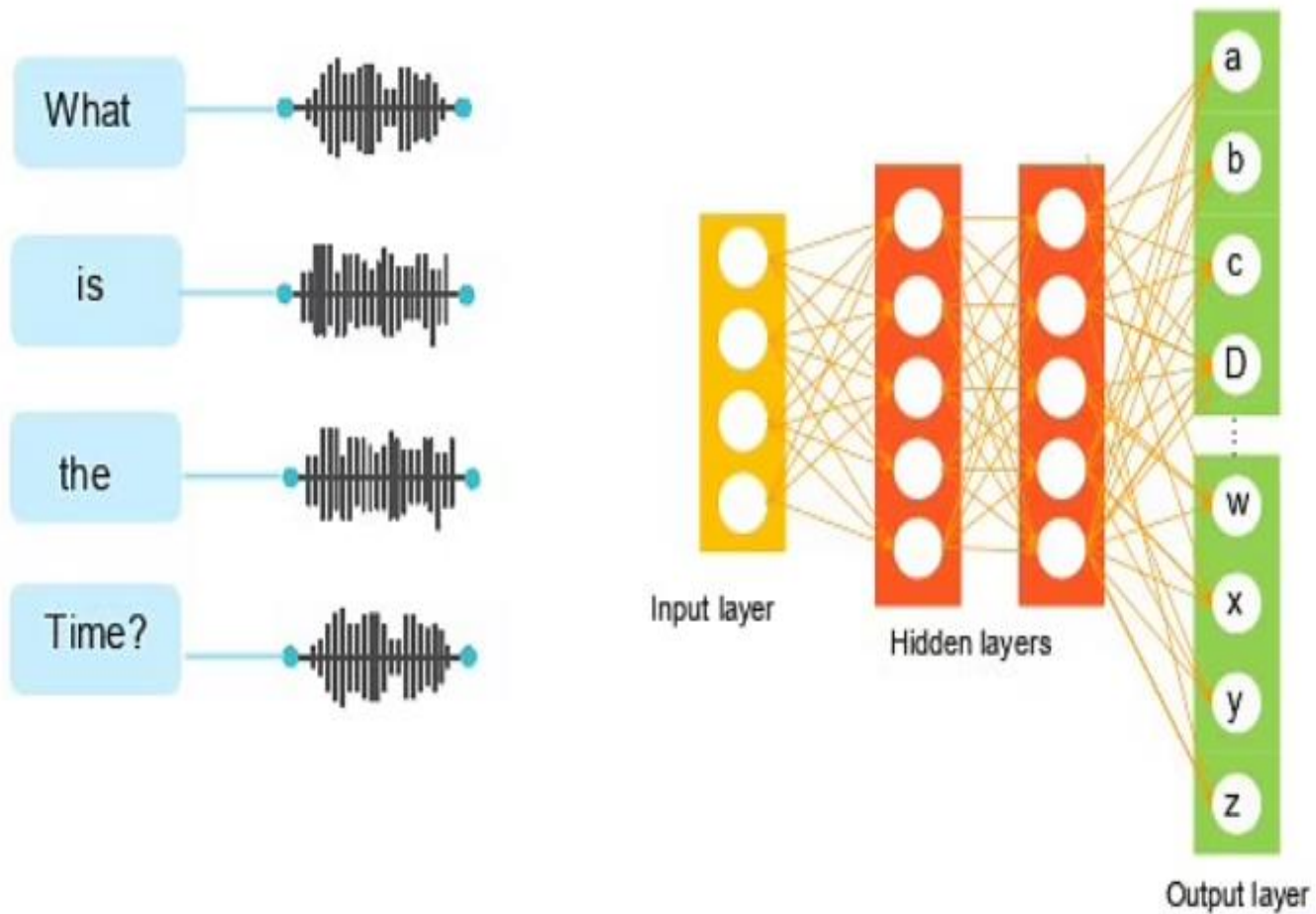
Rule-based NLP vs. Statistical NLP:

- Natural Language Processing is separated in two different approaches:
- **Rule-based Natural Language Processing:**
 - It uses common sense reasoning for processing tasks. For instance, the freezing temperature can lead to death, or hot coffee can burn people's skin, along with other common sense reasoning tasks. However, this process can take much time, and it requires manual effort.
- **Statistical Natural Language Processing:**
 - It uses large amounts of data and tries to derive conclusions from it. Statistical NLP uses machine learning algorithms to train NLP models. After successful training on large amounts of data, the trained model will have positive outcomes with deduction.

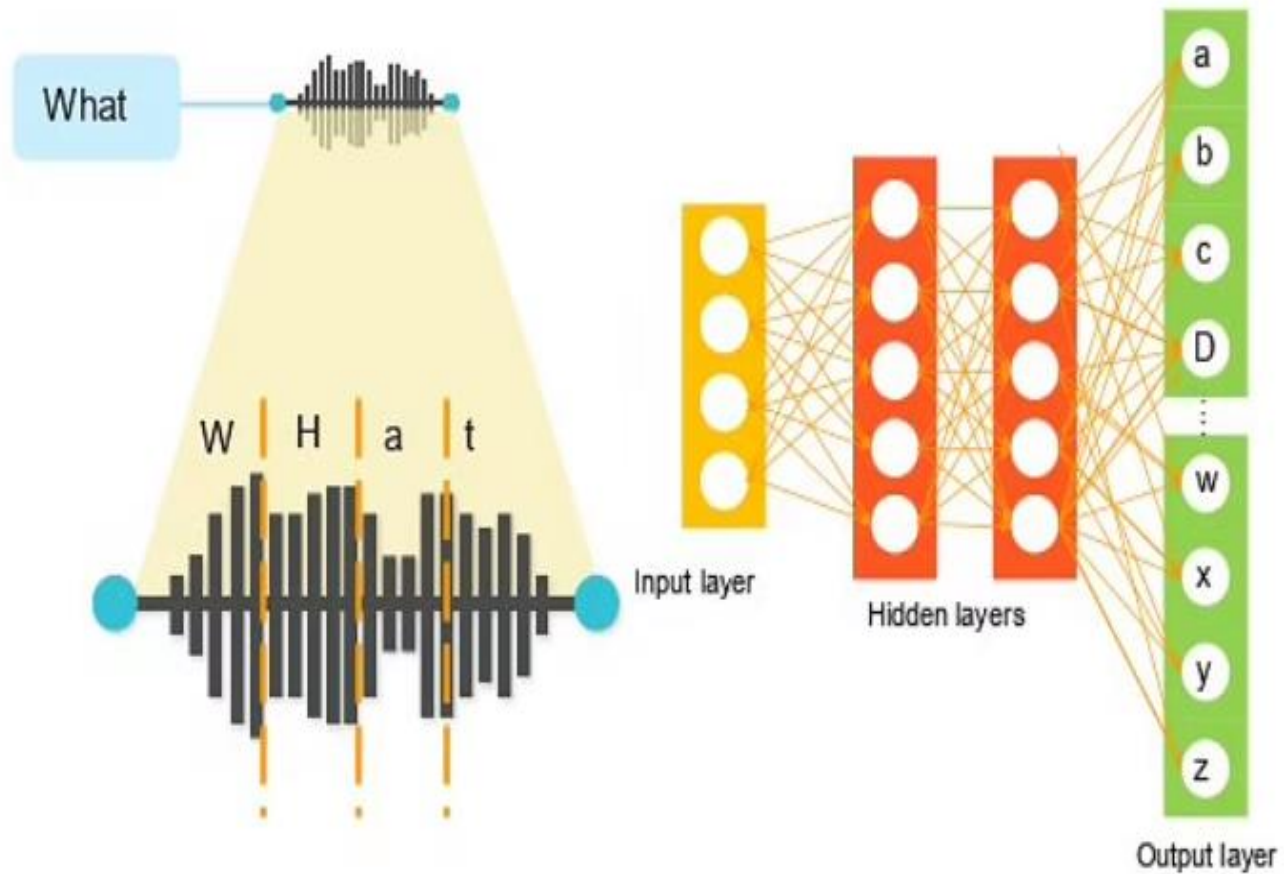
Rule-based NLP vs. Statistical NLP:

Rule Based NLP	Statistical NLP
+ 1) Flexible	+ 1) Easy to scale
+ 2) Easy to debug	+ 2) Learn by itself
+ 3) Doesn't require much training	+ 3) Fast development
+ 4) Understanding language	+ 4) High coverage
+ 5) High precision	
- 1) Requires skilled developers	- 1) Requires large amount of data
- 2) Slow parsing	- 2) Difficult to debug
- 3) Moderate coverage	- 3) No understanding of context

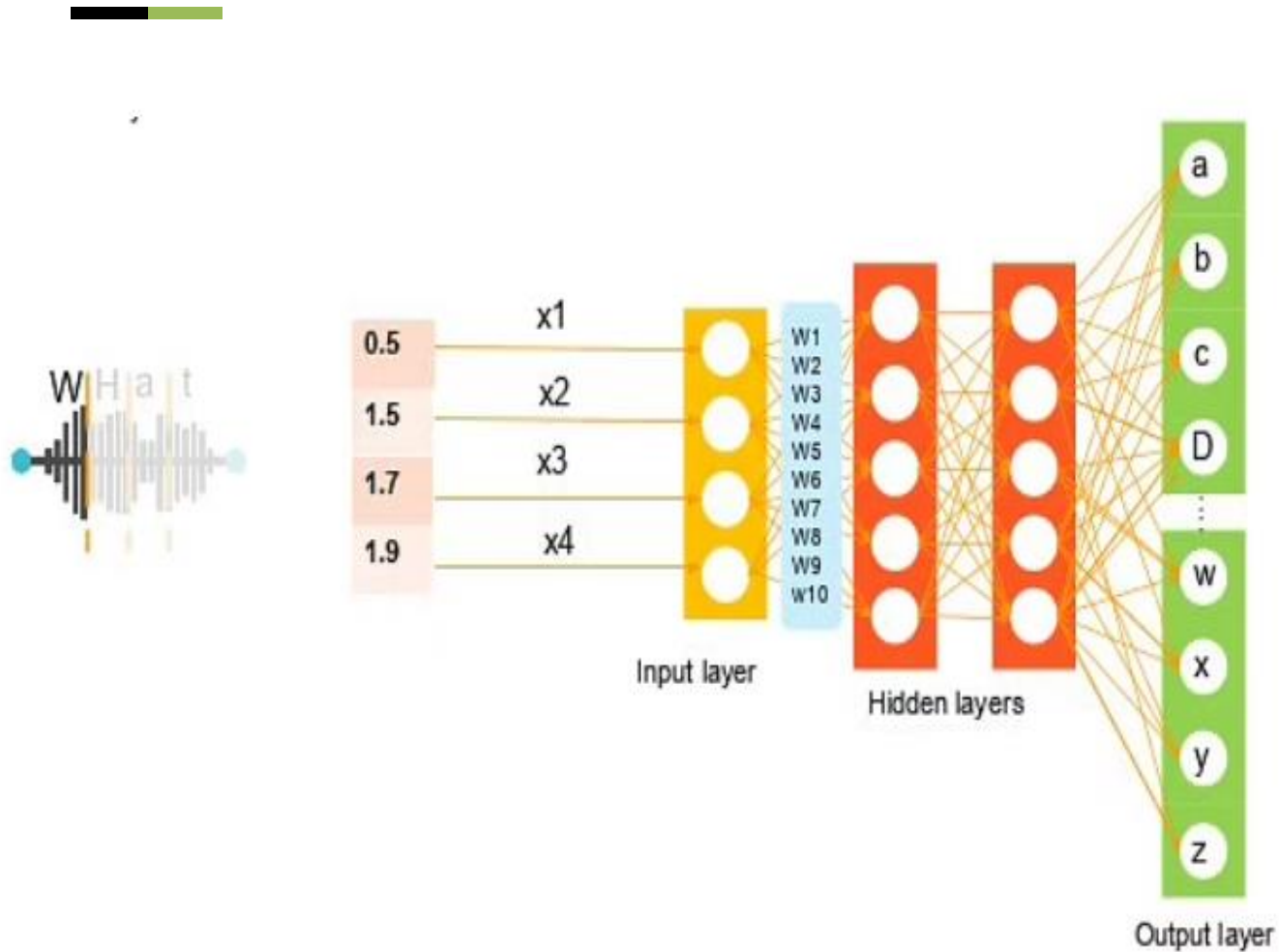
Example:



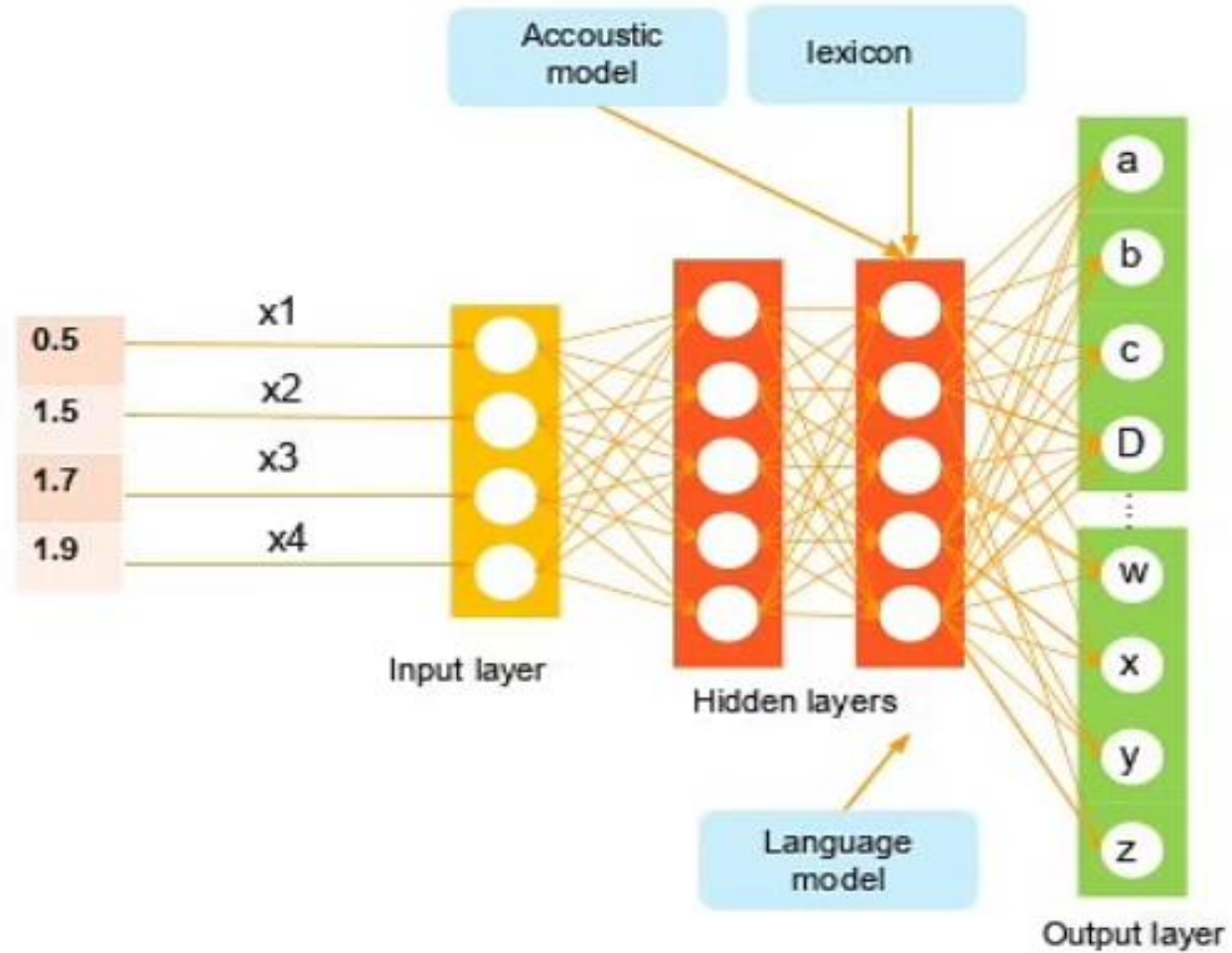
Example:



Example:



model contains a large list of
id the probability of them
occurring



NLP Process

- End-to-end steps in Natural Language Processing.
 - 1) Data acquisition – corpus
 - 2) Data visualization – Text visualization
 - 3) Text pre-processing- tokenization (white-space tokenization, regular-expression tokenization, stemming), Noise Removal – Stop Words
 - 4) Feature Engineering – text vectorization(Binary term frequency, Bag of Words, L1, L2, Word2Vec)
 - 5) Model building and evaluation

Data acquisition – corpus

- A **Corpus** is defined as a collection of text documents
- For example a data set containing news is a corpus or the tweets containing Twitter data is a corpus
- Corpus consists of documents, documents comprise paragraphs, paragraphs comprise sentences and sentences comprise further smaller units which are called **Tokens**.

Corpus : Example

```
corpus = ["This is a brown house. This house is big. The street number  
is 1.",  
          "This is a small house. This house has 1 bedroom. The street  
number is 12.",  
          "This dog is brown. This dog likes to play.",  
          "The dog is in the bedroom."]
```

Text pre-processing- Tokenization

- Tokenization – process of converting a text into tokens
- Tokens – words or entities present in the text
- Text object – a sentence or a phrase or a word or an article

Text pre-processing- Tokenization

- Tokens can be words, phrases, or Engrams, and **Engrams** are defined as the group of n words together.
- For example, consider this given sentence-
“I love my phone.”
- uni-grams($n=1$) are: I, love, my, phone
- Di-grams($n=2$) are: I love, love my, my phone
- And tri-grams($n=3$) are: I love my, love my phone

Text pre-processing- Tokenization

- White-space Tokenization
- In this process, the entire text is split into words by splitting them from white spaces.
- For example, in a sentence- “I went to New-York to play football.”
- This will be splitted into following tokens: “I”, “went”, “to”, “New-York”, “to”, “play”, “football.”

Text pre-processing- Tokenization

- Regular Expression Tokenization - in which a regular expression pattern is used to get the tokens.
- For example, consider the following string containing multiple delimiters such as comma, semi-colon, and white space.
- Sentence= "Football, Cricket; Golf Tennis"
- Tokens= "Football", "Cricket", "Golf", "Tennis"
- **Tokenization can be performed at the sentence level or at the word level or even at the character level.**

Text pre-processing- Tokenization

- Stemming
- Stemming is an elementary rule-based process for removing inflectional forms from a token and the outputs are the **stem** of the word.
- For example, “laughing”, “laughed”, “laughs”, “laugh” will all become “laugh”, which is their stem, because their inflection form will be removed.
- For example, consider a sentence: “His teams are not winning”
- After stemming the tokens that we will get are- “hi”, “team”, “are”, “not”, “winn”

Noise Entities Removal

- Noise is considered as that piece of text which is not relevant to the context of the data and the final output.
- Stop words are considered as the noise in the text.
- Stop words usually refer to the most common words such as “and”, “the”, “a” in a language.
- **For Example,**
- Language stop words (commonly used words of a language – is, am, the, of, in, etc),
- URLs or links,
- Social media entities (mentions, hashtags),
- Punctuations, and Industry-Specific words.

Noise Entities Removal

- The general steps which we have to follow to deal with noise removal are as follows:
 - Firstly, prepare a dictionary of noisy entities,
 - Then, iterate the text object by tokens (or by words),
 - Finally, eliminating those tokens which are present in the noise dictionary.

Text Visualization

- Text visualization is the technique of using graphs, charts, or [word clouds](#) to showcase written data in a visual manner. This provides quick insight into the most relevant keywords in a text, summarizes content, and reveals trends and patterns across documents.

Text Visualization

- **Summarize large amounts of text.** Automatically highlight key terms in a series of texts, and categorize text by topic, sentiment, and more, saving hours of reading time. How long would it take you to read 500 online reviews? With a word cloud or data visualization dashboard, you can understand text data at a glance.
- **Make text data easy to understand.** The human brain loves visual data. In fact, we are able to process images much faster than text. Text visualization is an effective way of simplifying complex data and communicating ideas and concepts to team managers.

Text Visualization

- **Find insights in qualitative data.** Customer feedback holds a trove of insights. Through text visualization, you can get an overview of the features, products, and topics that are most important to your customers. Learn what their pain points are and what you're doing right.
- **Discover hidden trends and patterns.** Analyze and visualize insights over time to detect fluctuations, and quickly find the root cause.

Text Visualization : Word Cloud

- Word Cloud is a data visualization technique in which words from a given text display on the main chart. Some properties associated with this chart are as follows:
- In this technique, more frequent or essential words display in a larger and bolder font,
- While less frequent or essential words display in smaller or thinner fonts.
- This data visualization technique gives us a glance at what text should be analyzed, so it is a very beneficial technique in NLP tasks.

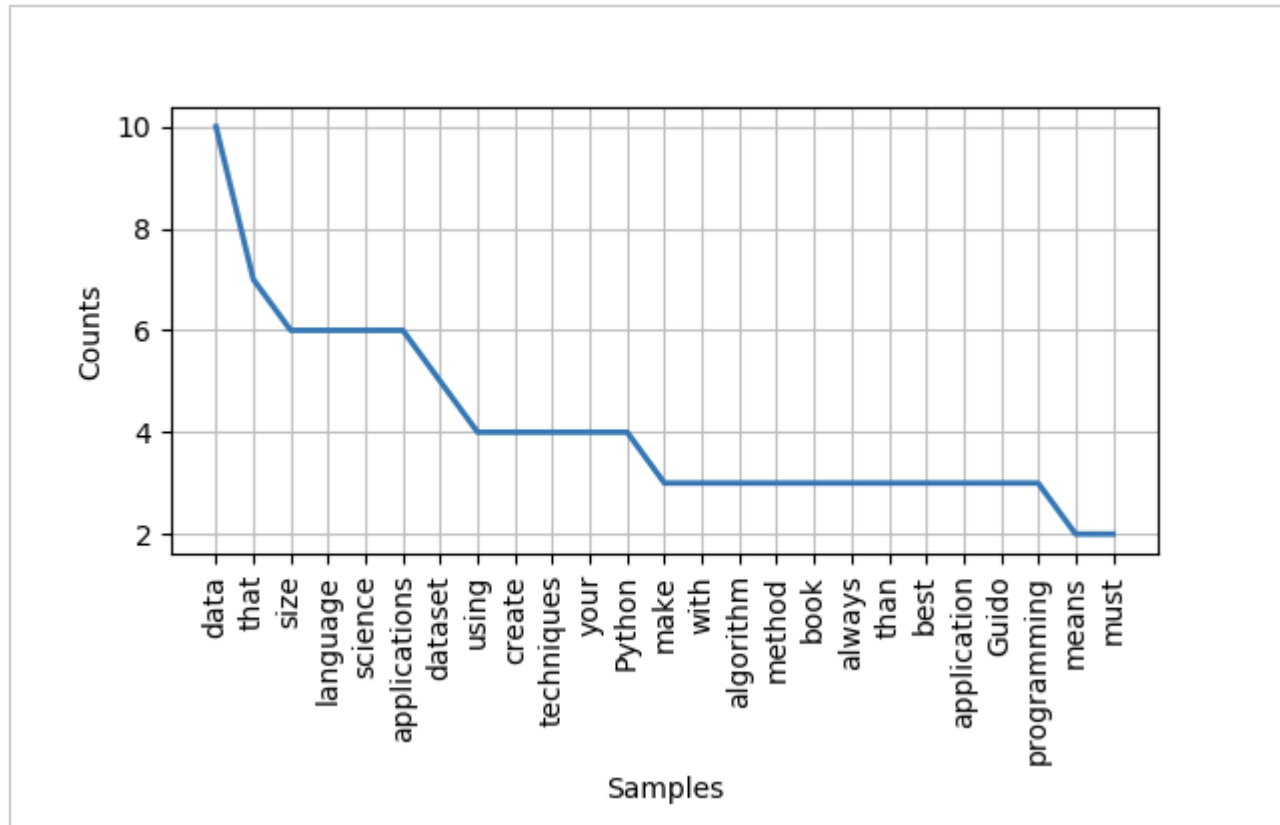
Text Visualization : Word Cloud

- **Advantages of Word Cloud**
 - They are fast.
 - They are engaging.
 - They are simple to understand.
 - They are casual and visually appealing.
- **Disadvantages of Word Cloud**
 - They are non-perfect for non-clean data.
 - They lack the context of words.

Text Visualization : Frequency Distribution

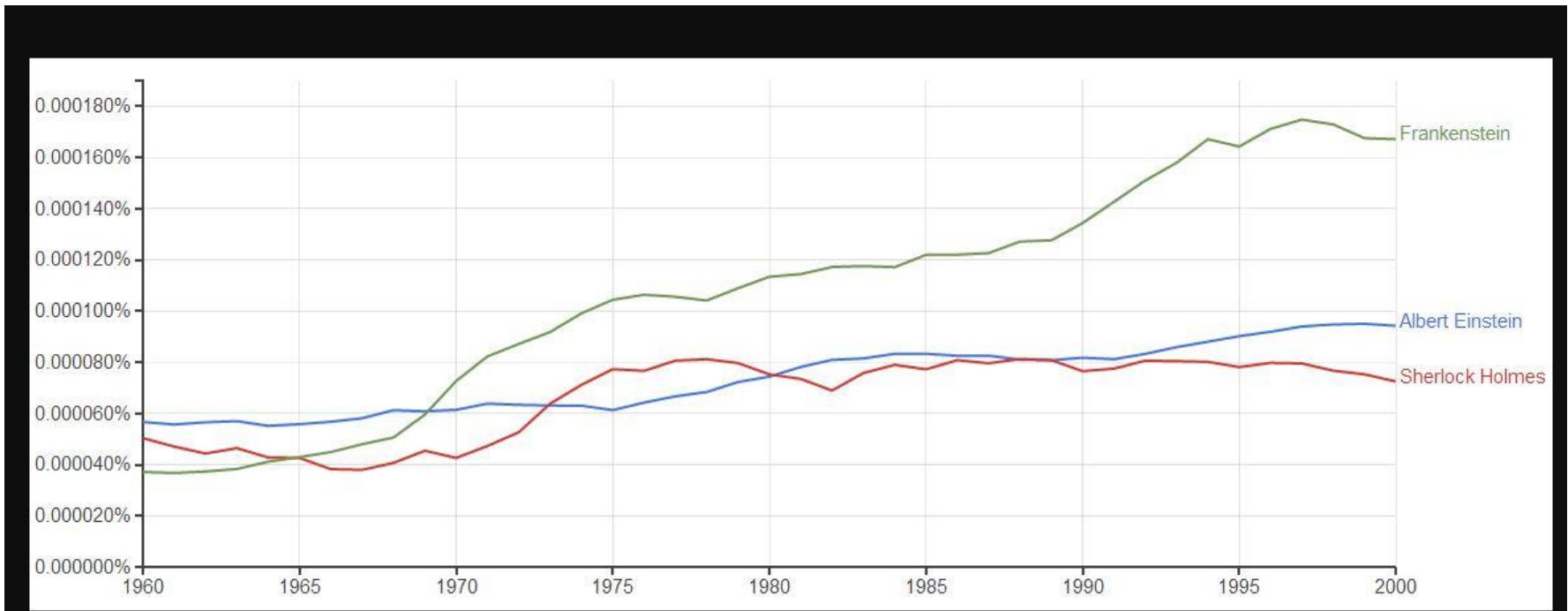
- A frequency distribution records the number of times each outcome of an experiment has occurred. For example, a frequency distribution could be used to record the frequency of each word type in a document. Formally, a frequency distribution can be defined as a function mapping from each sample to the number of times that sample occurred as an outcome.
- Frequency distributions are generally constructed by running a number of experiments, and incrementing the count for a sample every time it is an outcome of an experiment.

Text Visualization : Frequency Distribution



Text Visualization

- A graph generated by Google Ngram Viewer with key words “Frankenstein” “Albert Einstein” and “Sherlock Holmes” to show the development of them.*



Feature Engg. : Text Vectorization

- ***Text Vectorization*** is the process of converting text into numerical representation.
- Some popular methods to accomplish text vectorization:
- Binary Term Frequency
- Bag of Words (BoW) Term Frequency
- (L1) Normalized Term Frequency
- (L2) Normalized TF-IDF
- Word2Vec

Text Vectorization : Binary Term Frequency

- Binary Term Frequency captures presence (1) or absence (0) of term in document.

document. For this part, under TfidfVectorizer, we set binary parameter equal to

Saikat Chandra 11 Towards

	12	bedroom	big	brown	dog	house	likes	number	play	small	street
0	0.0	0.0	1.0	1.0	0.0	1.0	0.0	1.0	0.0	0.0	1.0
1	1.0	1.0	0.0	0.0	0.0	1.0	0.0	1.0	0.0	1.0	1.0
2	0.0	0.0	0.0	1.0	1.0	0.0	1.0	0.0	1.0	0.0	0.0
3	0.0	1.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0

Text Vectorization : Bag of Words (BoW) Term Frequency

- It is a method of extracting essential features from raw text so that we can use it for machine learning models.
- We call it “**Bag**” of words because we discard the order of occurrences of words.
- A bag of words model converts the raw text into words, and it also counts the frequency for the words in the text.
- In summary, a bag of words is a collection of words that represent a sentence along with the word count where the order of occurrences is not relevant.

Text Vectorization : BoW Process

1. Raw Text: This is the original text on which we want to perform analysis.
2. Clean Text: Since our raw text contains some unnecessary data like punctuation marks and stopwords, so we need to clean up our text. Clean text is the text after removing such words.
3. Tokenize: Tokenization represents the sentence as a group of tokens or words.
4. Building Vocab: It contains total words used in the text after removing unnecessary data.
5. Generate Vocab: It contains the words along with their frequencies in the sentences.

Text Vectorization : BoW Process

Example:

1. Jim and Pam traveled by bus.
2. The train was late.
3. The flight was full. Traveling by flight is expensive.

a. Creating a basic structure:

Sentence 1	Sentence2	Sentence 3
Jim	The	The
and	train	flight
Pam	was	was
travelled	late	full
by		Travelling
the		by
bus		flight
		is
		expensive

b. Words with frequencies:

Sentence1	Count	Sentence2	Count	Sentence3	Count
Jim	1	The	1	The	1
and	1	train	1	flight	2
Pam	1	was	1	was	1
travelled	1	late	1	full	1
by	1			Travelling	1
the	1			by	1
bus	1			is	1
				expensive	1

Text Vectorization : BoW Process

Example:

1. Jim and Pam traveled by bus.
2. The train was late.
3. The flight was full. Traveling by flight is expensive.

c. Combining all the words:

Sentence	Frequency
and	1
bus	1
by	2
expensive	1
flight	2
full	1
is	1
jim	1
late	1
pam	1
the	3
train	1
travelled	1
travelling	1
was	1

d. Final model:

	and	bus	by	expensive	flight	full	is	jim	Late	pam	The	train	travelled	travelling	was
S-1	1	1	1	0	0	0	0	1	0	1	1	0	1	0	0
S-2	0	0	0	0	0	0	0	0	1	0	1	1	0	0	1
S-3	0	0	1	1	2	1	1	0	0	0	1	0	0	1	1

Text Vectorization : BoW

- Bag of Words (BoW) Term Frequency captures frequency of term in document

	12	bedroom	big	brown	dog	house	likes	number	play	small	street
0	0.0	0.0	1.0	1.0	0.0	2.0	0.0	1.0	0.0	0.0	1.0
1	1.0	1.0	0.0	0.0	0.0	2.0	0.0	1.0	0.0	1.0	1.0
2	0.0	0.0	0.0	1.0	2.0	0.0	1.0	0.0	1.0	0.0	0.0
3	0.0	1.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0

Text Vectorization : BoW

- Limitations:
 1. Semantic meaning: It does not consider the semantic meaning of a word. It ignores the context in which the word is used.
 2. Vector size: For large documents, the vector size increase, which may result in higher computational time.
 3. Preprocessing: In preprocessing, we need to perform data cleansing before using it.

Text Vectorization : (L1) Normalized Term Frequency

- (L1) Normalized Term Frequency captures normalized BoW term frequency in document.

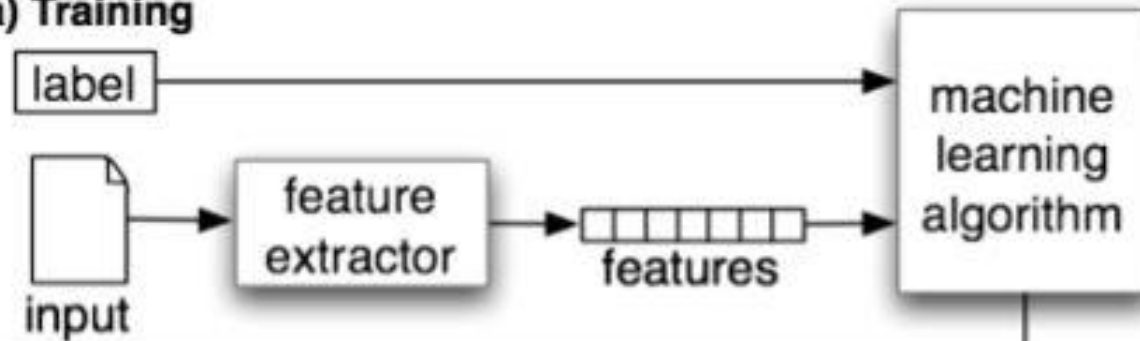
	12	bedroom	big	brown	dog	house	likes	number	play	small	street
0	0.000000	0.000000	0.166667	0.166667	0.0	0.333333	0.0	0.166667	0.0	0.000000	0.166667
1	0.142857	0.142857	0.000000	0.000000	0.0	0.285714	0.0	0.142857	0.0	0.142857	0.142857
2	0.000000	0.000000	0.000000	0.200000	0.4	0.000000	0.2	0.000000	0.2	0.000000	0.000000
3	0.000000	0.500000	0.000000	0.000000	0.5	0.000000	0.0	0.000000	0.0	0.000000	0.000000

Text Classification

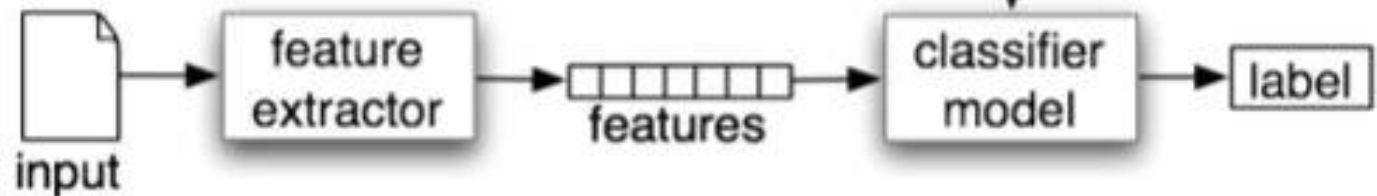
- Text classification is one of the classical problem of NLP.
- Examples include – Email Spam Identification, topic classification of news, sentiment classification and organization of web pages by search engines.
- Text classification, in common words is defined as a technique to systematically classify a text object (document or sentence) in one of the fixed category.
- It is really helpful when the amount of data is too large, especially for organizing, information filtering, and storage purposes.
- A typical natural language classifier consists of two parts: (a) Training (b) Prediction.
- Firstly the text input is processes and features are created. The machine learning models then learn these features and is used for predicting against the new text.

Text Classification

(a) Training



(b) Prediction



Text Classification

```
training_corpus = [  
    ('I am exhausted of this work.', 'Class_B'),  
    ("I can't cooperate with this", 'Class_B'),  
    ('He is my badest enemy!', 'Class_B'),  
    ('My management is poor.', 'Class_B'),  
    ('I love this burger.', 'Class_A'),  
    ('This is an brilliant place!', 'Class_A'),  
    ('I feel very good about these dates.', 'Class_A'),  
    ('This is my best work.', 'Class_A'),  
    ("What an awesome view", 'Class_A'),  
    ('I do not like this dish', 'Class_B')]  
  
test_corpus = [  
    ("I am not feeling well today.", 'Class_B'),  
    ("I feel brilliant!", 'Class_A'),  
    ('Gary is a friend of mine.', 'Class_A'),  
    ("I can't believe I'm doing this.", 'Class B'),
```

Text Classification

```
model = NBC(training_corpus)
print(model.classify("Their codes are amazing. "))
>>> "Class_A"
print(model.classify("I don't like their computer. "))
>>> "Class_B"
print(model.accuracy(test_corpus))
>>> 0.83
```

Text Matching / Similarity : Clustering

- One of the important areas of NLP is the matching of text objects to find similarities.
- Important applications of text matching includes automatic spelling correction, data de-duplication and genome analysis etc.
- A number of text matching techniques are available depending upon the requirement.

Text Similarity : Levenshtein Distance

- Levenshtein Distance – The Levenshtein distance between two strings is defined as the minimum number of edits needed to transform one string into the other, with the allowable edit operations being insertion, deletion, or substitution of a single character. Following is the implementation for efficient memory computations.
- `levenshtein("analyze","analyse")`

Text Similarity : Phonetic Matching

- Phonetic Matching – A Phonetic matching algorithm takes a keyword as input (person's name, location name etc) and produces a character string that identifies a set of words that are (roughly) phonetically similar.
- It is very useful for searching large text corpuses, correcting spelling errors and matching relevant names.
- Soundex and Metaphone are two main phonetic algorithms used for this purpose.
- Python's module Fuzzy is used to compute soundex strings for different words,

Text Similarity : Phonetic Matching

-

```
import fuzzy
soundex = fuzzy.Soundex(4)
print soundex('ankit')
>>> "A523"
print soundex('aunkit')
>>> "A523"
...
```

Text Similarity : Jaccard

- There are various text similarity metric exist such as Cosine similarity, Euclidean distance and Jaccard Similarity. All these metrics have their own specification to measure the similarity between two queries.
- Jaccard Similarity matrix used to determine the similarity between two text document means how the two text documents close to each other in terms of their context that is how many common words are exist over total words.
- The Jaccard Similarity score is in a range of 0 to 1. If the two documents are identical, Jaccard Similarity is 1. The Jaccard similarity score is 0 if there are no common words between two documents.

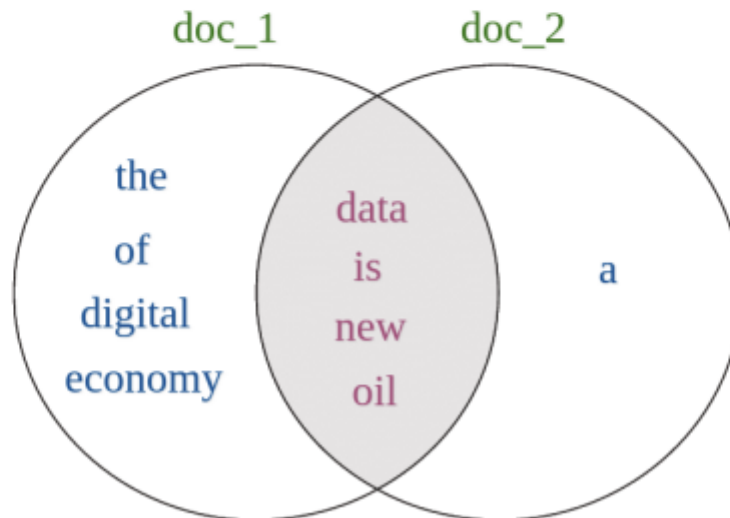
Text Similarity : Jaccard

```
doc_1 = "Data is the new oil of the digital economy"  
doc_2 = "Data is a new oil"
```

```
words_doc1 = {'data', 'is', 'the', 'new', 'oil', 'of', 'digital', 'economy'}  
words_doc2 = {'data', 'is', 'a', 'new', 'oil'}
```

Text Similarity : Jaccard

$$\begin{aligned} J(doc_1, doc_2) &= \frac{\{'data', 'is', 'the', 'new', 'oil', 'of', 'digital', 'economy'\} \cap \{'data', 'is', 'a', 'new', 'oil'\}}{\{'data', 'is', 'the', 'new', 'oil', 'of', 'digital', 'economy'\} \cup \{'data', 'is', 'a', 'new', 'oil'\}} \\ &= \frac{\{'data', 'is', 'new', 'oil'\}}{\{'data', 'a', 'of', 'is', 'economy', 'the', 'new', 'digital', 'oil'\}} \\ &= \frac{4}{9} = 0.444 \end{aligned}$$



Text Similarity : Cosine Clustering

- Cosine Similarity – When the text is represented as vector notation, a general cosine similarity can also be applied in order to measure vectorized similarity.
- Cosine similarity is one of the metric to measure the text-similarity between two documents irrespective of their size in Natural language Processing. A word is represented into a vector form. The text documents are represented in n-dimensional vector space.
- Mathematically, Cosine similarity metric measures the cosine of the angle between two n-dimensional vectors projected in a multi-dimensional space. The Cosine similarity of two documents will range from 0 to 1. If the Cosine similarity score is 1, it means two vectors have the same orientation. The value closer to 0 indicates that the two documents have less similarity.

Text Similarity : Cosine Clustering

```
doc_1 = "Data is the oil of the digital economy"  
doc_2 = "Data is a new oil"
```

```
# Vector representation of the document
```

```
doc_1_vector = [1, 1, 1, 1, 0, 1, 1, 2]
```

```
doc_2_vector = [1, 0, 0, 1, 1, 0, 1, 0]
```

	data	digital	economy	is	new	of	oil	the
doc_1	1	1	1	1	0	1	1	2
doc_2	1	0	0	1	1	0	1	0

$$A \cdot B = \sum_{i=1}^n A_i B_i$$

$$= (1 * 1) + (1 * 0) + (1 * 0) + (1 * 1) + (0 * 1) + (1 * 0) + (1 * 1) + (2 * 0)$$

$$= 3$$

$$\sqrt{\sum_{i=1}^n A_i^2} = \sqrt{1 + 1 + 1 + 1 + 0 + 1 + 1 + 4} = \sqrt{10}$$

$$\sqrt{\sum_{i=1}^n B_i^2} = \sqrt{1 + 0 + 0 + 1 + 1 + 0 + 1 + 0} = \sqrt{4}$$

$$\text{cosine similarity} = \cos\theta = \frac{A \cdot B}{|A||B|} = \frac{3}{\sqrt{10} * \sqrt{4}} = 0.4743$$

Important Libraries for NLP (python)

- Scikit-learn: Machine learning in Python
- Natural Language Toolkit (NLTK): The complete toolkit for all NLP techniques.
- Pattern – A web mining module for the with tools for NLP and machine learning.
- TextBlob – Easy to use nlp tools API, built on top of NLTK and Pattern.
- spaCy – Industrial strength NLP with Python and Cython.
- Gensim – Topic Modelling for Humans
- Stanford Core NLP – NLP services and packages by Stanford NLP Group.